ChIP-Seq: technical considerations for obtaining high-quality data

Benjamin L Kidder, Gangqing Hu & Keji Zhao

Chromatin immunoprecipitation followed by next-generation sequencing analysis (ChIP-Seq) is a powerful method with which to investigate the genome-wide distribution of chromatin-binding proteins and histone modifications in any genome with a known sequence. The application of this technique to a variety of developmental and differentiation systems has provided global views of the *cis*-regulatory elements, transcription factor function and epigenetic processes involved in the control of gene transcription. Here we describe several technical aspects of the ChIP-Seq assay that diminish bias and background noise and allow the consistent generation of high-quality data.

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Immune function involves the tight control of transcription by the interplay of *cis*-regulatory elements and *trans*-acting transcription factors; this is influenced mainly by the epigenetic landscape of cells of the immune response. Published studies have identified many transcription factors and chromatin modifiers linked to innate and adaptive immunity. However, traditionally only one or a few individual gene(s) or regulatory region(s) have been studied by techniques such as chromatin immunoprecipitation (ChIP)-PCR assays. Technical advances in global techniques such as DNA microarray and next-generation sequencing have allowed the analysis of entire genomes. The identification and characterization of the genome-wide locations of transcription factors and chromatin-modifying enzymes and the modification status of histones is imperative for comprehensive understanding of the transcriptional regulation of the immune system under diverse biological conditions. The application of ChIP followed by next-generation sequencing analysis (ChIP-Seq) to several transcription factors and epigenetic modifications has propelled efforts to characterize their global cistromes (*cis*-acting targets of a *trans*-acting factor across the genome) and to elucidate immune memory[1–4]. Here we highlight several technical aspects of ChIP-Seq that should be considered for the generation of high-quality genome-wide data, including antibodies, controls, library construction and statistical analysis.

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Antibodies

The quality of antibodies used for ChIP-Seq is one of the most important factors that contribute to the quality of the data generated by these studies. Antibodies that offer high sensitivity and specificity are necessary for ChIP-Seq because they allow the detection of enrichment peaks without substantial background noise. Many commercial antibodies that have been tested for their use in ChIP studies are available. However, results from various groups have shown that not all commercial antibodies that are designated as 'ChIP grade' or 'ChIP qualified' can be successfully used to investigate genome-wide protein-DNA interactions. Certain antibodies that are sufficient for the detection of locus-specific enrichment by ChIP-PCR may not be suitable for ChIP-Seq studies. As a general rule, if an antibody identifies enrichment of fivefold of more in ChIP-PCR assays at several positive-control regions relative to enrichment at negative control regions, it usually works well for ChIP-Seq. Because enrichment may vary from target to target, enrichment at multiple genomic loci should be assessed after ChIP.

It is also important to consider the potential cross-reactivity of antibodies with closely related family members that may serve unique or redundant roles in the immune system. The specificity of an antibody can be directly addressed by immunoblot analysis of a protein of interest through the use of models of knockdown by RNA-mediated interference or genetic knockout. In these cases, because expression of the protein should be diminished to background amounts, any protein detected by immunoblot analysis can be assumed to be nonspecific. ChIP with a higher concentration of antibody, which can be acquired from various companies by request, or pooled monoclonal antibodies, may also be considered to enrich for factor-occupied DNA sequences.

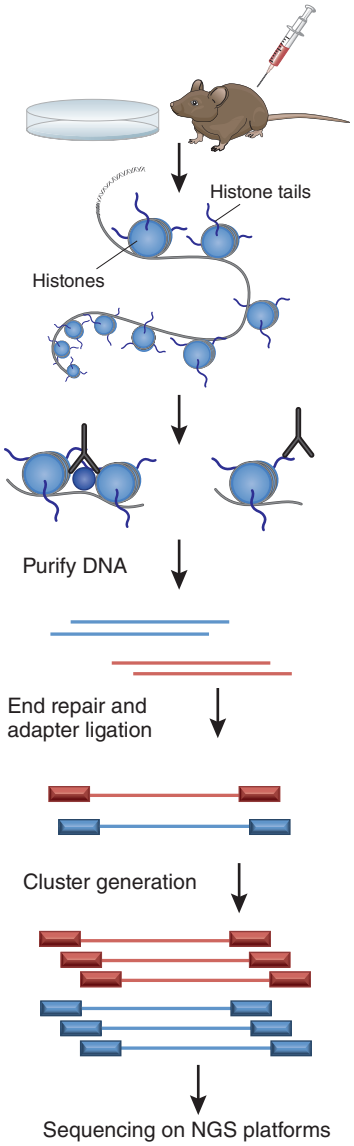
In cases in which specific antibodies are unavailable, epitope-tagged proteins can be expressed, and then a tag-specific antibody can be used for ChIP[5,6]. The most frequently used tags include hemagglutinin, Flag, Myc and V5. Although this method has been successful for certain applications, the efficiency of these tags in ChIP varies depending on the specific protein they fuse to and the fusion location in the protein (amino or carboxyl terminus). In addition to being tagged with epitope antibodies, the target protein can also be tagged with a biotin acceptor sequence, which can be labeled with biotin via biotin ligase either *in vivo* or *in vitro*. The high affinity of the biotin-streptavidin interaction can withstand stringent wash conditions and thus results in much lower background noise[6,7]. This strategy is particularly advantageous when partially denaturing conditions are needed to expose

epitopes such as components of large protein complexes. One caveat to this approach is that overexpression of proteins may lead to altered genomic binding profiles due to excess protein in the cell. Therefore, it is important to ensure that the expression of the tagged protein does not exceed the endogenous expression of the protein.

The clonality, or heterogeneity, of the antibody should also be considered in the selection of an antibody. Monoclonal antibodies recognize a single epitope on an antigen, which may be beneficial for diminishing background noise in ChIP studies. However, the use of monoclonal antibodies may result in a lower signal if the epitope is masked by surrounding chromatin components or if the protein is part of a larger protein complex. Although epitope recognition may be problematic for any antibody in ChIP studies regardless of its clonality, polyclonal antibodies offer the flexibility of the recognition of multiple epitopes, which may boost the signal in cases in which epitopes are masked by surrounding material. Because there is no definitive rule for choosing the appropriate clonality of an antibody for ChIP studies, it is best to test several antibodies, if available. This will provide greater confidence that the peaks identified are true positive results.

Cell number

The abundance of the protein or histone modification to be investigated and the quality of the antibody should be considered when determining the number of cells to begin ChIP-Seq analysis. As the signal-to-noise ratio is directly correlated with the cell number, the use of more cells tends to produce a higher signal-to-noise ratio. Therefore, it is important to empirically determine the minimum number of cells that can be used, whenever possible. ChIP-Seq experiments typically require 1×10^6 to 10×10^6 cells, which results in 10–100 ng of ChIP DNA (**Fig. 1**). The former (1×10^6 cells) is usually sufficient for the analysis of abundant proteins such as RNA polymerase II and localized histone modifications such as trimethylation of histone H3 at Lys4 (H3K4me3), whereas the latter (10×10^6 cells) may be required for the analysis of less-abundant proteins or diffuse histone modifications. However, several alternative protocols have been designed that require fewer cells (1×10^4 to 1×10^5) for the profiling of genome-wide distributions of histone modifications[8], although these methods have not been demonstrated yet to work well for transcription factors. The advantage of this approach is the ability to use 1–10% as many as cells as are used in conventional ChIP-seq protocols, which may be beneficial for the study of rare cell types.



Isolation of cells of the immune response

- Use the correct number of cells: 1×10^6 to 10×10^6
- Collect biological replicates of cells
- Choose an appropriate control for antibody specificity (knockout or RNAi knockdown)

Fragmentation by sonication or MNase treatment

- Shear chromatin to a size range of ~150–300 bp
- Sonicate chromatin extracts for non-histone proteins
  - * Sonication conditions should be determined empirically for each cell type
- Treat chromatin extracts with MNase for analysis of histone modifications
  - * Do not overdigest chromatin

ChIP analysis of histone modifications, transcription factors or epigenetic regulators

- Select antibody: monoclonal versus polyclonal
- Choose reference control (Input or IgG)
- Perform ChIP with established protocols
- Purify DNA

Library construction

- Do end repair and adapter ligation
- Perform PCR using primers compatible with sequencing platform
  - * Avoid overamplifying DNA

Sequencing

- Determine sequencing depth on the basis of the prevalence of binding throughout the genome: more sequencing tags may be needed for diffuse signals (such as H3K27me3)
- Perform single-end or paired-end sequencing

Figure 1 ChIP-Seq experimental design. ChIP-Seq is a powerful tool with which to investigate protein-DNA interactions on a global scale. It is important that the appropriate controls for antibody specificity be determined before ChIP-Seq is begun. After isolation of the ideal number of cells, chromatin is sheared into an ideal size range by sonication or enzymatic means (micrococcal nuclease (MNase)). Next, high-quality antibodies are used for ChIP to enrich for factor-occupied DNA sequences. After purification of ChIP-enriched DNA, a library is constructed to allow sequencing on next-generation sequencing (NGS) platforms. Library construction typically includes end-repair, the addition of single adenosine residues, adaptor ligation and PCR with primers compatible with the sequencing platform. After cluster generation, single- or paired-end sequencing is performed on next-generation sequencing platforms. RNAi, RNA-mediated interference; bp, base pairs.

Controls

An important part of ChIP-Seq experimental design is determining which controls to use (**Fig. 1**). Artifacts may arise in the following steps of experimentation. First, in terms of chromatin fragmentation, open chromatin regions are easier to shear than are closed chromatin regions and thus the former may be associated with higher background signals[9]. Second, antibodies may have unrecognized cross-reactivity. Third, there may be variable sequencing efficiency of DNA regions with different base compositions. Although both nonspecific immunoglobulin G (IgG) antibodies and input chromatin have been used as controls, IgG may be less desirable in certain circumstances because of the following reasons: most IgG antibodies are not obtained from true preimmune serum from the same animal in which the specific antibody was raised; and IgG antibodies usually immunoprecipitate much less DNA than specific antibodies do, and thus limited genomic regions

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from the control may be overamplified during the library construction step. In this case, the resulting sequence 'reads' will not cover the genome as sufficiently as a background model would for peak identification. Therefore, input chromatin serves as a better control for bias in chromatin fragmentation and variations in sequencing efficiency; additionally, it provides greater and more evenly distributed coverage of the genome. However, normal IgG antibodies and input chromatin are not the appropriate controls for addressing the cross-reactivity of antibodies, which can be controlled for through the use of true preimmune serum or a different antibody that is specific for the same factor but recognizes a different epitope. Additional controls for antibody specificity include targeted deletion or knockdown of the factor of interest by RNA-mediated interference. Many mice are available with knockout of transcription factors that have important regulatory functions in the immune system (such as GATA-3, STAT4, STAT5, STAT6 and so on); these would serve as ideal controls for testing antibody specificity. In these cases, because expression of the protein should be diminished to background amounts, any potential DNA-binding events can be assumed to be nonspecific.

Replicates

High-quality ChIP-Seq data sets are valuable resources for the community. Many factors, including cell-culture conditions, ChIP and library construction, may contribute to variability between data sets. To ensure reliability of the data, biological replicate experiments are necessary. Although there is no consensus on the correct number of replicates needed, at least duplicate biological experiments should be done. Although only one ChIP-grade antibody is available for the analysis of most histone modifications and transcription factors, it is recommended that ChIP-Seq data be confirmed through the use of a different antibody wherever possible to control for potential antibody cross-reactivity.

Chromatin fragmentation

Before ChIP, chromatin must be fragmented into a manageable size (~150–300 base pairs) by sonication or enzymatic means (usually via treatment with micrococcal nuclease) with or without crosslinking, depending on the purpose of the experiment (**Fig. 1**). For histone modification, digestion of native chromatin into mononucleosome-sized particles with micrococcal nuclease may be the preferred method because this generates high-resolution data for nucleosome modifications and eliminates signal artifacts caused by crosslinking with other genomic regions. However, this method may potentially suffer loss of signal due to unstable nucleosomes. For mapping of binding sites for transcription factors, sonication of formaldehyde-crosslinked chromatin may be the preferred method because micrococcal nuclease degrades linker DNA, where transcription factors tend to bind. Although chromatin fragments of different sizes may work well for ChIP-PCR assays, the optimal size range of chromatin for ChIP-Seq analysis should be between 150 and 300 base pairs. DNA fragments in this size range, which are equivalent to mono- and dinucleosome chromatin fragments, provide high-resolution analysis of binding sites, and they work well for next-generation sequencing platforms. The conditions used to sonicate chromatin need to be optimized for each cell type because they are highly variable and depend on the cell type, the number of cells used, fixation conditions, type of sonicator and sonicator settings. It is important to avoid oversonication of chromatin when transcription factors are to be evaluated by ChIP, whereas oversonication may not be as problematic for analysis of histone modifications. Preparing nuclei before fixation may also help diminish background signals, which may be observed with whole-cell chromatin extracts.

Sonication buffers may also influence ChIP-Seq results. Sonication in SDS-containing buffers may disrupt protein-protein and protein-DNA interactions and therefore may expose antibody epitopes buried inside a protein complex and improve specific signals. For example, efficient mapping of the methylation of H3K79, which is located in the nucleosome core, requires sonication of chromatin in SDS-containing buffers[10]. SDS-containing buffers also increase sonication efficiency and may be appropriate for the evaluation of transcription factors, which are tightly bound to DNA. However, the addition of SDS to sonication buffers may result in loss of signal for proteins not directly bound to DNA, such as epigenetic regulators.

Library construction and sequencing

Libraries may be constructed from ChIP DNA by standard protocols specific to the sequencing platform (**Fig. 1**). Typically, library construction includes end repair, the addition of single adenosine residues, adaptor ligation, size selection and gel purification, followed by PCR with primers specific to the sequencing platform. During the size-selection step, it is important that the agarose gel be melted at room temperature (~22 °C) rather than at 50 °C, as the latter temperature might result in a bias for guanosine and cytidine because of loss of sequences rich in adenosine and thymidine[11]. During the PCR amplification step, it is important that adaptor-ligated DNA products are not overamplified, which may result in a loss of specific signal, bias or redundancy in the number of sequencing tags. Overamplification can typically be avoided by decreasing the number of PCR cycles or decreasing the amount of template DNA used for PCR. One way to determine whether overamplification has occurred is comparison of the size of the adaptor-ligated product to that of the PCR product. Overamplified PCR products will generally have more of a shift in size than will adaptor-ligated products (for example, an adaptor-ligated product 200–400 base pairs in length may shift to a size of >300–500 base pairs).

Although the exact modifications of adapters and primers sold by Illumina and other providers of next-generation sequencing reagents are not publicly available, adapters and PCR primers can be custom-ordered from other companies to lower costs. For example, custom PCR primers with a phosphorothioate modification between the two bases at the end have produced results similar to those obtained with primers from Illumina, results for some investigators[11].

The quantity of PCR products is usually sufficient for sequencing as long as the product bands are visible on an agarose gel after staining with ethidium bromide. Libraries can be sequenced at a distance of 25 base pairs from one end of the DNA templates, which typically provides relatively good coverage (66%)[12] of uniquely mappable sequences in the human genome (build hg19 of the Genome Reference Consortium). Longer sequencing reads of 30 or 35 base pairs improve the mappability to 70.9% or 74.1%, respectively[12], and may be preferred if cost is not an issue.

Libraries can be sequenced by either a single-end sequencing strategy (which generates short sequence reads from one end of the DNA template) or a paired-end sequencing strategy (which generates short sequence reads from both ends of the DNA template; **Fig. 1**). Although both methods have been successful, paired-end sequencing has the following advantages: more sequencing coverage; improved efficiency of alignment to repetitive regions because more sequence information is obtained from each DNA template; and greater ability to detect fragment sizes. In cases in which ChIP-enriched DNA fragments partially overlap or contain repetitive sequences, sequencing of both ends may allow more accurate mapping to the genome than does single-end sequencing, which may otherwise result in a loss of repetitive sequences during the analysis.

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The number of sequencing reads required for reasonable genomic coverage is contingent on several factors, including antibody affinity and the number of target sites in the genome. For analysis of the distribution of histone H3 trimethylated at Lys4 (H3K4me3), 5×10^6 sequence reads are sufficient to reach saturation of its target sites, whereas 20×10^6 reads may be required for reasonable coverage for H3K27me3 profiles. A more quantitative approach for determining the appropriate depth of sequencing involves evaluating the saturation point (the number of reads after which additional sequencing does not identify new binding or enrichment sites)[13]. One lane of sequencing on a Genome Analyzer IIA, which typically generates around 20×10^6 mapped unique reads, is usually sufficient for analysis of most modifications and transcription factors. To lower the cost of sequencing, it is also possible to pool several libraries, such as for analysis of H3K4me3 modifications, through the use of indexing adaptors.

Data analysis

Vendor-supplied analysis packages include an image-analysis tool that transforms pixel values into intensities and a 'base-calling' tool that converts those intensities into sequences. Because the output sequences are short in length and high in error rates, third-party base-calling tools have been developed to increase base-call accuracy and yield[14]. Alternative tools correct for potential errors after base calling[15] and facilitate genome alignment and *de novo* assembly[16]. However, the potential benefit of such tools to peak calling for ChIP-Seq data remains unexplored. Short sequence reads, with or without error corrections, are then mapped to a reference genome through the use of a variety of alignment programs. Because of the development of alignment tools, short-read alignment is no longer a bottleneck in the data-analysis process[17].

The quality of ChIP-Seq data can be assessed by a combination of methods (**Fig. 2**). First, it is important that the summary report generated by the vendor-supplied analysis pipeline be evaluated. For example, the 'Summary.html' file from the CASAVA (consensus assessment of sequence and variation) data-analysis software program contains a set of comprehensive performance measures for data generated by Illumina genome analyzer platforms. The next step involves the conversion of sequence alignments to an appropriate format, followed by uploading of the results to the UCSC Genome Browser and examination of the genomic regions of interest (such as known targets of a transcription factor). Another qualitative measure for determining the quality of ChIP-Seq data involves searching for sequence motifs in tag-enriched regions or peaks[18]. In addition, it may be useful to examine the distribution of tag profiles around certain genomic features (such as transcriptional start sites). We also suggest that the inspection be made parallel through the use of input or IgG controls and that bias by minimized through the use of specific tools[19]. As mentioned above, it is also important that the identity of selected ChIP-Seq peaks be confirmed through the use of ChIP followed by quantitative PCR.

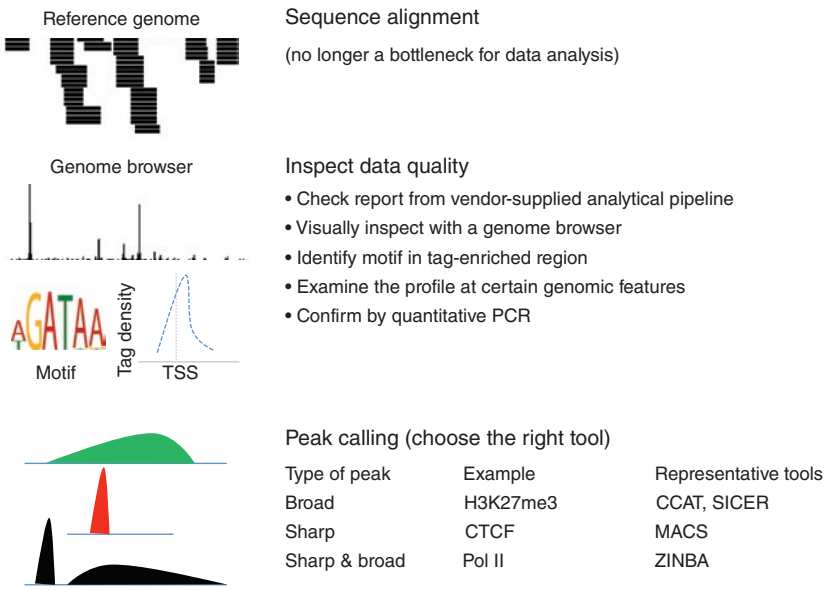

Figure 2 Common procedures for ChIP-seq data analysis. After base-calling, short-read sequences are aligned to a reference genome. Data quality is assessed by a combination of various strategies, such as visual inspection with a genome browser, motif identification and confirmation by ChIP and quantitative PCR. The initial inspection or prior knowledge provides information about whether the peaks are broad or sharp or both. Various algorithms (bottom right) have been developed for the identification of peaks of these three groups.

After the initial quality inspection, peak calling is done to identify tag-enriched regions from the ChIP-Seq data. Multiple algorithms are available, and comparisons of these constitute the theme of several publications[18]. When considering which tool to choose, it is important to recognize that there are two fundamental types of peaks: sharp and broad (**Fig. 2**). The inspection of tag distributions in a genome browser, together with prior knowledge, helps to define what the peaks look like. Algorithms such as MACS (model-based analysis of ChIP-Seq data)[20] work well for the identification of the sharp peaks of most sequence-specific transcription factors, whereas programs such as SICER (spatial clustering approach for the identification of ChIP-enriched regions)[21] and CCAT (control-based ChIP-Seq analysis tool)[22] are appropriate for the identification of the broad peaks of most histone modifications and chromatin-binding proteins. Our experience indicates that CCAT has greater sensitivity for identifying peaks, whereas SICER has greater specificity. However, because CCAT requires negative controls for the estimation of noise rates, this algorithm may not be applicable to data sets from which negative controls are absent, such as formaldehyde-assisted isolation of regulatory elements followed by sequencing. Another method, ZINBA (zero-inflated negative binomial algorithm), has been developed for the identification of both sharp and broad peaks[23]. Because these tools are designed for different purposes, comparison of their performance may not be fair. However, efforts evaluate these methods have been limited mainly because of the absence of objective benchmark standards[18].

Reads mapped to multiple sites ('multi-reads') are usually discarded during 'normal' analysis. Consequently, peaks in highly repetitive regions are overlooked. However, repetitive regions have been linked to important biological functions such as disease susceptibility, immunity and defense. A new method has been proposed to incorporate multi-reads into peak detection through the use of a weighted alignment scheme[24]. As most previously unknown peaks reside in repetitive regions, this method will be of particular use for the analysis of ChIP-Seq data for proteins that selectively bind to repetitive regions.

Another important issue in data analysis is comparison of the amount of histone modi-

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fication or binding of transcription factors in two different cell types or under different conditions. Because of variations in ChIP conditions, the amount of noise may vary substantially between different samples even with the same antibody. Because scaling the data to sequence depth does not eliminate systematic errors, normalization algorithms are needed for comparisons across samples. The tool DIME (differential identification using mixtures ensemble) has been developed for the classification of regions with considerable enrichment in one ChIP-Seq sample relative to their abundance in another ChIP-Seq sample based on an estimation of multivariate mixture models[25]. Because the method partitions the genome into bins that are larger in size than that of a typical transcription factor–binding site, it may serve as a first-pass algorithm with which to identify candidate regions for differences in binding, especially in cases of low sequencing depth. A sub-module of SICER is also able to identify regions with differences in enrichment in two conditions when regions of interest are specified (for example, the difference between two samples in their abundance of histone modifications at promoters).

Because genome-wide data are being generated at a rapid rate, it is important that analytical tools are developed at a similar rate to support the storage and analysis needs that users encounter. We expect a flurry of software with user-friendly interfaces (graphical user interfaces) to be released in the near future and adopted by biologists with diverse backgrounds.

Summary

Global surveys of sequence-specific binding of transcription factors and chromatin structure in a variety of cell types and species have provided comprehensive information with which to understand dynamic processes such as stem-cell differentiation, the formation of immune memory, disease progression and responses to external environmental stimuli. ChIP-Seq is a powerful technology with which to evaluate protein-DNA interactions on a global scale. The findings obtained by ChIP-Seq have improved the knowledge of the epigenetic landscapes that regulate chromatin structure and transcriptional profiles. Because data sets generated by ChIP-Seq studies are valuable resources for the community, it would be beneficial if data sets generated from different laboratories could be compared directly. For this purpose, experimental conditions must be optimized and standardized.

In this commentary we have discussed several important technical considerations for ChIP-Seq experiments and data analysis (**Figs. 1** and **2**) that may help yield reliable and reproducible data sets. As ChIP-Seq protocols are further improved and data-analysis platforms become more manageable for experimental biologists, the application of this technology will help build comprehensive global views of immune processes, thus aiding the understanding of normal development and disease states.

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COMPETING FINANCIAL INTERESTS
The authors declare no competing financial interests.

1. Barski, A. *et al. Cell* **129**, 823–837 (2007).
2. Wei, G. *et al. Immunity* **30**, 155–167 (2009).
3. Northrup, D.L. & Zhao, K. *Immunity* **34**, 830–842 (2011).
4. Natoli, G. *Immunity* **33**, 12–24 (2010).
5. Goldberg, A.D. *et al. Cell* **140**, 678–691 (2010).
6. Kolodziej, K.E. *et al. BMC Mol. Biol.* **10**, 6 (2009).
7. Kim, J., Chu, J., Shen, X., Wang, J. & Orkin, S.H. *Cell* **132**, 1049–1061 (2008).
8. Adli, M., Zhu, J. & Bernstein, B.E. *Nat. Methods* **7**, 615–618 (2010).
9. Teytelman, L. *et al. PLoS ONE* **4**, e6700 (2009).
10. Steger, D.J. *et al. Mol. Cell. Biol.* **28**, 2825–2839 (2008).
11. Quail, M.A. *et al. Nat. Methods* **5**, 1005–1010 (2008).
12. Koehler, R., Issac, H., Cloonan, N. & Grimmond, S.M. *Bioinformatics* **27**, 272–274 (2011).
13. Kharchenko, P.V., Tolstorukov, M.Y. & Park, P.J. *Nat. Biotechnol.* **26**, 1351–1359 (2008).
14. Ledergerber, C. & Dessimoz, C. *Brief Bioinform.* published online, doi:10.1093/bib/bbq077 (18 January 2011).
15. Kao, W.C., Chan, A.H. & Song, Y.S. *Genome Res.* **21**, 1181–1192 (2011).
16. Kelley, D.R., Schatz, M.C. & Salzberg, S.L. *Genome Biol.* **11**, R116 (2010).
17. Li, H. & Homer, N. *Brief. Bioinform.* **11**, 473–483 (2010).
18. Szalkowski, A.M. & Schmid, C.D. *Brief Bioinform.* published online, doi:10.1093/bib/bbq068 (8 November 2010).
19. Cheung, M.S., Down, T.A., Latorre, I. & Ahringer, J. *Nucleic Acids Res.* **39**, e103 (2011).
20. Zhang, Y. *et al. Genome Biol.* **9**, R137 (2008).
21. Zang, C. *et al. Bioinformatics* **25**, 1952–1958 (2009).
22. Xu, H. *et al. Bioinformatics* **26**, 1199–1204 (2010).
23. Rashid, N., Giresi, P.G., Ibrahim, J.G., Sun, W. & Lieb, J.D. *Genome Biol.* **12**, R67 (2011).
24. Chung, D. *et al. PLOS Comput. Biol.* **7**, e1002111 (2011).
25. Taslim, C., Huang, T. & Lin, S. *Bioinformatics* **27**, 1569–1570 (2011).

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